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UNIVERSITY OF TECHNOLOGY AND APPLIED SCIENCE – IBRI
MATHEMATICS FOR COMPUTING/MATHEMATIS FOR IT
MATH2101/MATH201. SELF-STUDY QUIZ

Max Marks 5
Max Time: 20 mins

Name:_____ **ID No:**_____

1) In a game of tossing two coins, a winning price of OMR 5 is given if two heads appear, OMR 2 for one head and OMR 0 for no heads. Create a distribution table and find the mean, variance, and standard distribution of winning. [2 Marks]

$S = \{\underbrace{HT}_{1}, \underbrace{TH}_{1}, \underbrace{HH}_{2}, \underbrace{TT}_{0}\}$

	0 OMR	2 OMR	5 OMR
x	0	2	5
$P(x)$	0.25	0.5	0.25
x^2	0	4	25

Mean $= \sum x\,(p(x))$

$= 0 \times 0.25 + 2 \times 0.5 + 5 \times 0.25$

$= 0 + 1 + 1.25$

$= 2.25$

Variance $= \sum x^2\, p(x) - \mu^2$

$= 0 \times 0.25 + 0.5 \times 4 + 25 \times 0.25$

$= 0 + 2 + 6.25 - 5.06$

$= 8.25 - 5.06$

$= 3.19$

Standard deviation $= \sqrt{\text{variance}}$

$= \sqrt{3.19}$

$= 1.78$

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2) A random variable X has the following probability distribution table: [3 Marks]

X	1	2	3	4	5	6
$P(X)$	$3k^2$ 0.12	$2k$ 0.4	k 0.2	0	$5k^2$ 0.2	$2k^2$ 0.08

$= 1$

Determine the following:

a. The value of $k = 0.2$

b. $P(X \geq 4) = 0.28$

c. $P(1 < X < 5) = 0.6$

$3k^2 + 2k + k + 0 + 5k^2 + 2k^2 = 1$

(A) $10k^2 + 3k - 1 =$

$\boxed{k = 0.2}$ $k = -0.5$

choose it

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(B) $p(x \geq 4)$

$0 + 0.2 + 0.08 = 0.28$

(C) $p(1 < x < 5)$

$0.4 + 0.2 + 0 = 0.6$